ECE 239AS.2

Google Cloud Compute Credits

Many thanks to CS PhD student Zhizheng Liu for developing this.

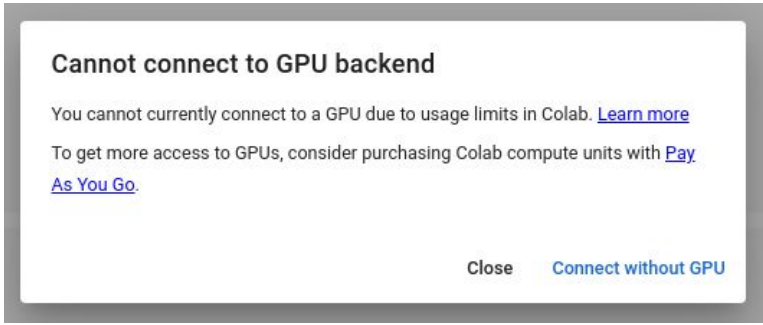
Discussion Goals

- Claim Google Cloud Credit
- Set up virtual machines (VMs) with GPUs using Google Cloud Credit
- Set up Google Colab to connect to VM as back-end

Claiming Google Cloud Credit

Google Cloud Credits - ECE 239AS

- Google has provided us Google cloud Education Credits worth $50 per coupon
- You can use Google Cloud VMs equipped with GPUs to run deep learning code, such as existing codebases available online, for your course project
- You can also use a VM as the back-end for your Google Colab – prevents your notebook from restarting randomly, and avoiding errors such as:



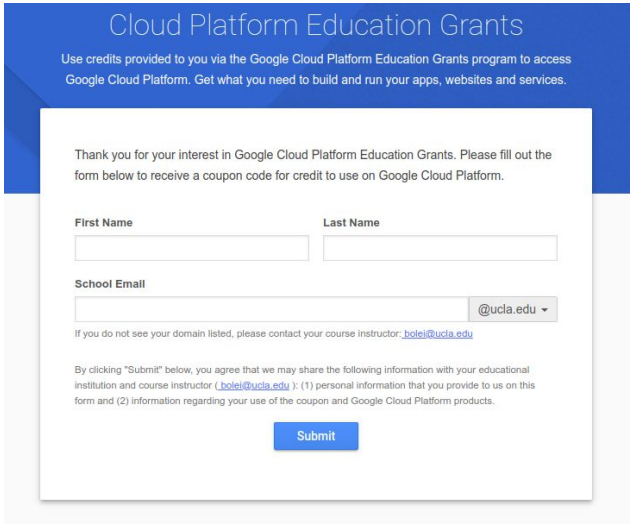
Important note on credit usage

- $50 seems like a lot of money, but in reality, this will probably last at most one week if you run your GPU VM continuously
- **Very important: Stop your VM when not in use to make your credits last longer!**

Retrieve your coupon using link below

Student Coupon Retrieval Link

https://gcp.secure.force.com/GCPEDU?cid=ZblLrbCA0sBNiXZqs7LYiS2tELS52sECjMRo2QFkDy3m5PdPm%2FKJ%2FcWps8Q6xd9s/



- You will be asked for a name and email address, which needs to match your school domain (@ucla.edu). A confirmation email will be sent to you with a coupon code.
- You can request a coupon from the URL and redeem it until: 7/31/2025
- Coupon valid through: 3/31/2026
- You can only request ONE code per unique email address.

Important note on claiming coupon

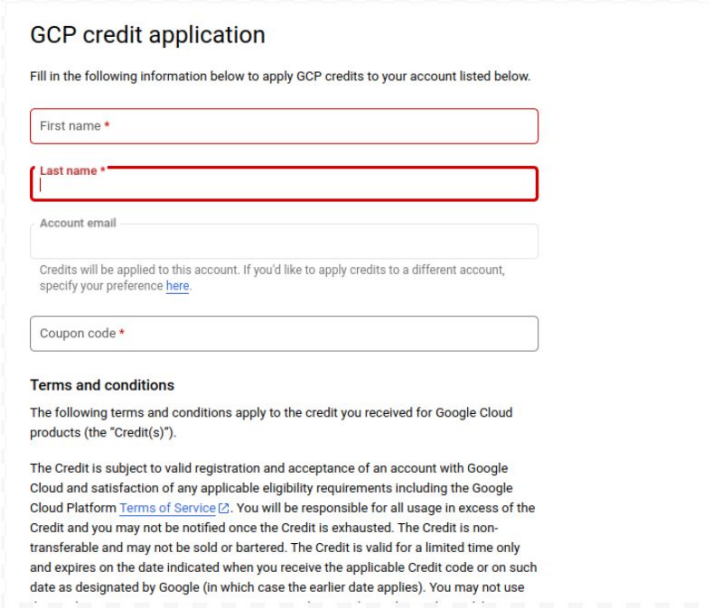
Please do not claim more than one credit coupon per person.

The number of coupons available is limited, only sufficient for 1 coupon per student. If you claim > 1 coupon, other students might not be able to claim any.

Redeeming Coupon

After submitting your coupon application, you will receive a link in your email with your unique coupon code.

Click on it and enter the necessary information



GCP credit application

Fill in the following information below to apply GCP credits to your account listed below.

First name *

Last name *

Account email

Credits will be applied to this account. If you'd like to apply credits to a different account, specify your preference here.

Coupon code *

Terms and conditions

The following terms and conditions apply to the credit you received for Google Cloud products (the "Credit(s)").

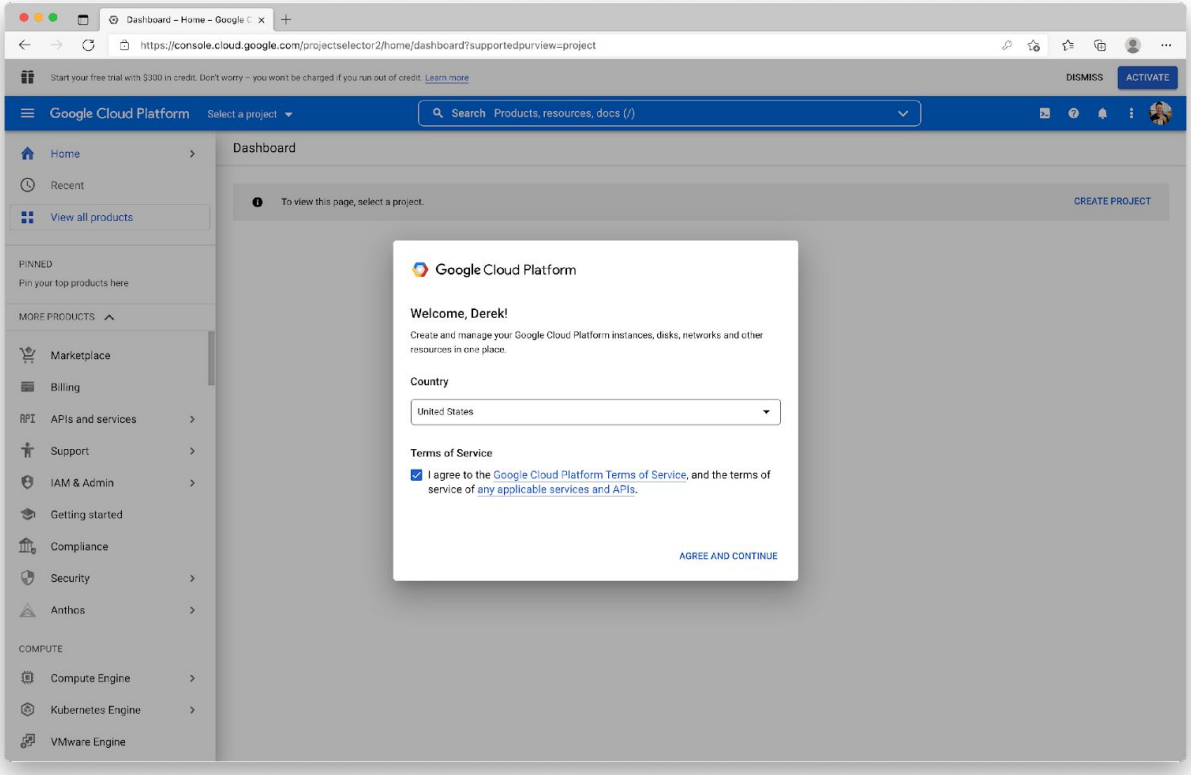
The Credit is subject to valid registration and acceptance of an account with Google Cloud and satisfaction of any applicable eligibility requirements including the Google Cloud Platform Terms of Service. You will be responsible for all usage in excess of the Credit and you may not be notified once the Credit is exhausted. The Credit is non-transferable and may not be sold or bartered. The Credit is valid for a limited time only and expires on the date indicated when you receive the applicable Credit code or on such date as designated by Google (in which case the earlier date applies). You may not use

Create Instance with GPU on GCP

Slides partially taken from: Mingyu Derek Ma from CS188: Natural Language Processing

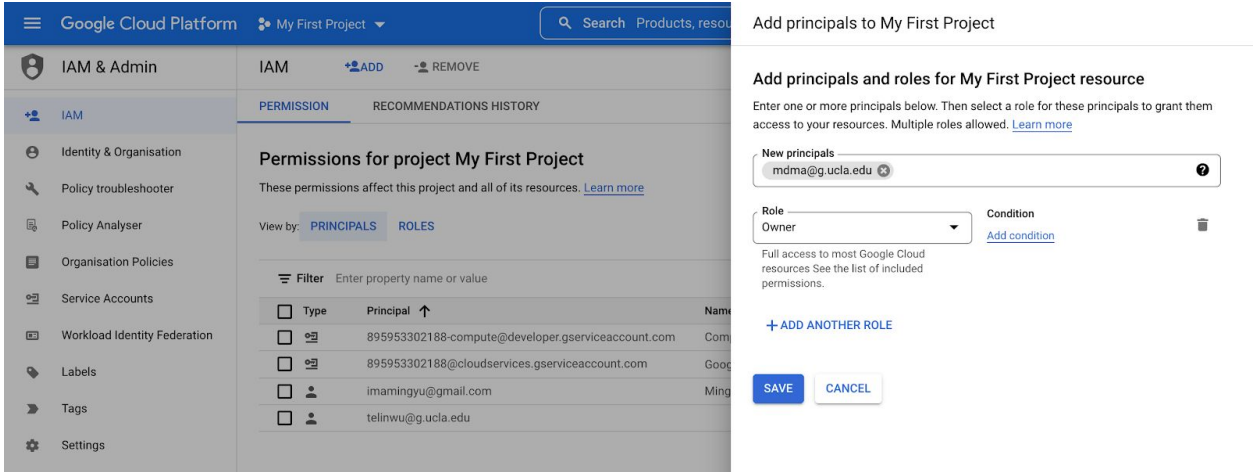
1. Create project

- Click ”Create Project”
- Or “New Project” after clicking the project name next to the “Google Cloud Platform” title
- New project may also be accessible after the organization (e.g. “g.ucla.edu”)



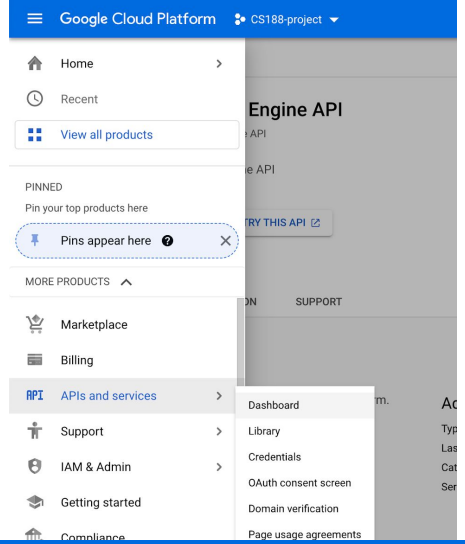
Share project with teammates

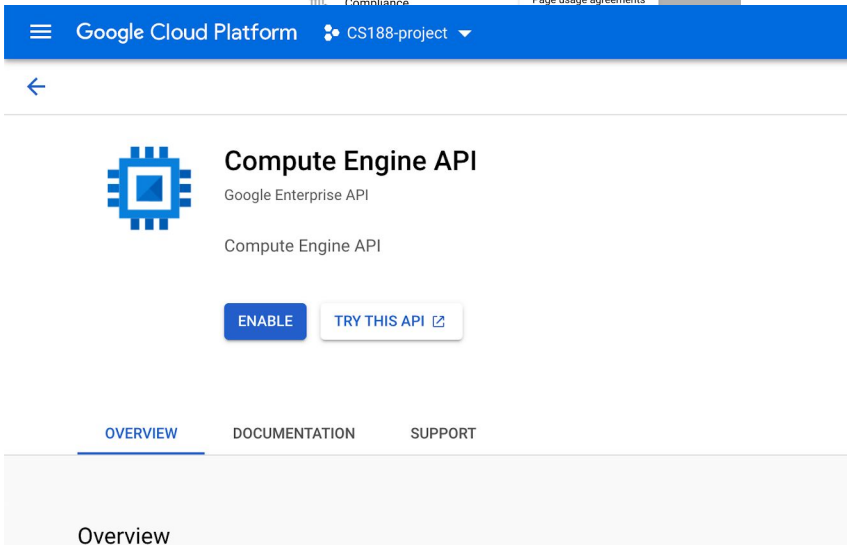
- *“IAM & Admin” > “IAM”*
- Add new user to the project, so other teammates can access the instances under this project



2. Enable Compute Engine API

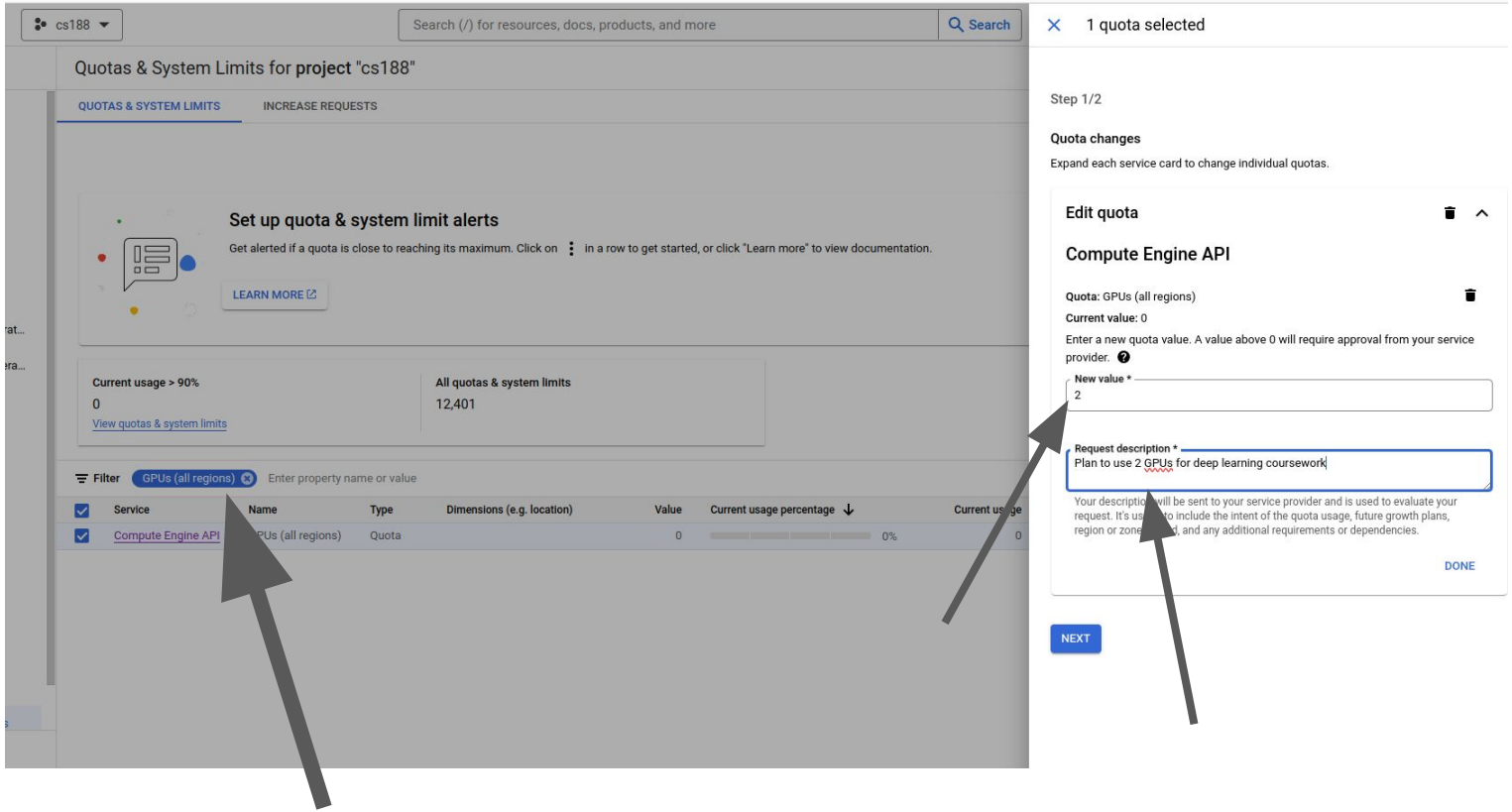
- It will prompt you to enable to the API when you first open the interfaces for Compute Engine
- Otherwise you can enable the API at *“API and services” > “Dashboard” > Search “Compute Engine API” > Enable*
- This can take several minutes



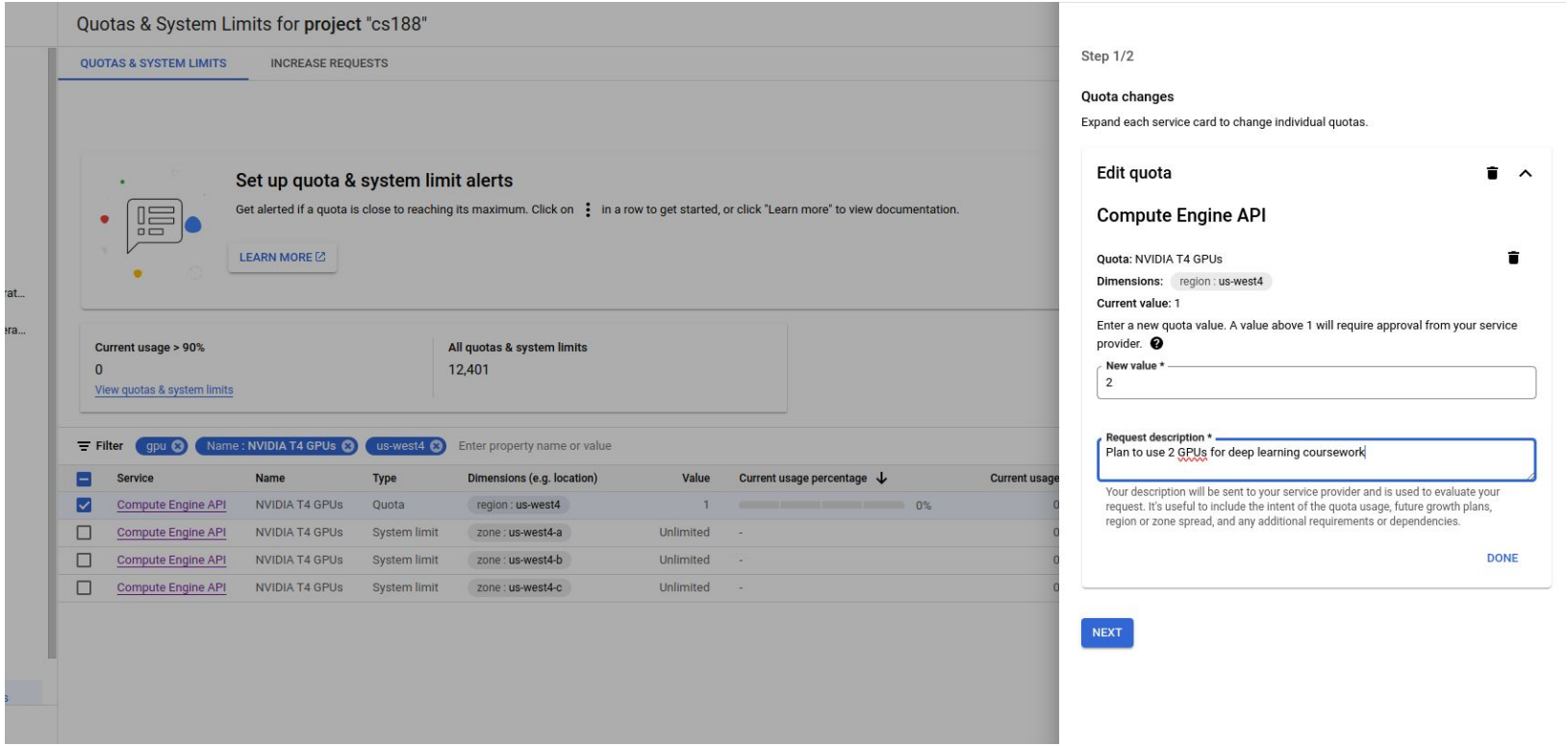


3. Check/change GPU quota after 48 hours

- By default, we can use 0 GPUs
- We need to request an increase in GPU quota
- Resource quotas | Compute Engine Documentation | Google Cloud
- Check quota at “IAM & Admin” > “Quotas”
- Add “GPUs (all regions)” in the filter
- Select quota item, click “Edit Quotas” (click the triple dots)
- Submit quota change request, need 24-48 hours to get response
  - Submit the quota increase request after 48 hours of creating your project, otherwise it will be declined
- us-west2 GPUs might already be available without making a quota request.



- Increase quota for *GPUs (all regions)*



- Increase quota for specific region and type of GPU you want to use (for example NVIDIA T4 GPUs at us-west4 is limited to 1 in the screenshot)

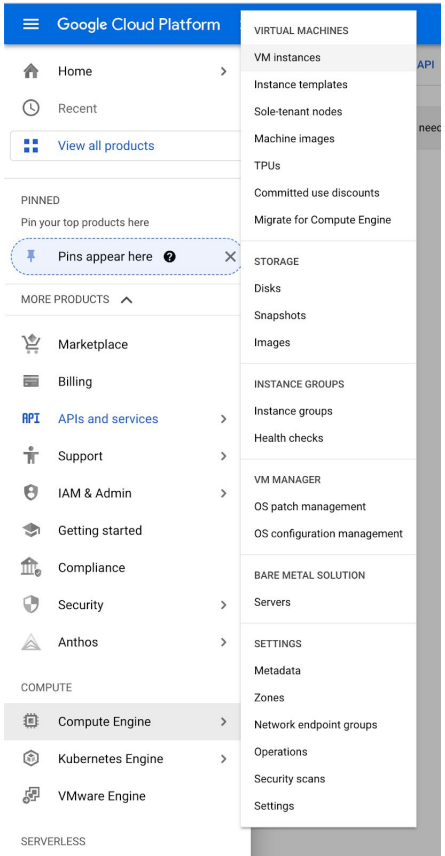
- [You can come back to this later after you have created your VM]

GPU Choices

- GPUs on Compute Engine | Compute Engine Documentation | Google Cloud
- GPUs pricing | Compute Engine: Virtual Machines (VMs) | Google Cloud
- GPU regions and zones availability | Compute Engine Documentation | Google Cloud

4. Create an instance with attached GPUs

- Enter “Compute Engine” > “VM Instances” > “Create Instance”



4. Create an instance with attached GPUs

- Create an instance
- Choose region and zone that has the GPU you requested
  - Check region supported GPU types in this link
  - For example, we choose “us-west4-b” to use NVIDIA T4 GPU
  - You must have an available quota in the region requested in order to create the instance
- Choose “GPU” under “Machine configuration”
- Select GPU type and number

Name *
instance-1

MANAGE TAGS AND LABELS

Region *
us-west4 (Las Vegas)
Region is permanent

Zone *
us-west4-b
Zone is permanent

Machine configuration

General purpose | Compute optimized | Memory optimized | GPUs

Graphics processing units (GPUs) accelerate specific workloads on your instances such as machine learning and data processing. Learn More

GPU type
NVIDIA T4

Number of GPUs
1

Enable Virtual Workstation (NVIDIA GRID)

	Series	Description	vCPUs	Memory	Platform
◉	N1	Balanced price & performance	1 - 96	1.8 - 624 GB	Intel Skylake

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads.
Or, you can create a custom machine for your workload's particular needs. Learn more

PRESET | CUSTOM

n1-standard-1 (1 vCPU, 3.75 GB memory)

vCPU	Memory
1	3.75 GB

ADVANCED CONFIGURATIONS

Monthly estimate

$217.51

That's about $0.30 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
1 vCPU + 3.75 GB memory	$39.05
1 NVIDIA T4	$270.10
10 GB balanced persistent disk	$1.10
Use discount	-$92.75
Total	**$217.51**

Compute Engine pricing

LESS

4. Create an instance with attached GPUs

Google Cloud | gcp-test0 | Search (/) for resources, docs, products, and more | Search

Create an instance | Create VM from...

- Machine configuration
  n1-standard-1 (1 NVIDIA T4), asia-east1-c
- OS and storage
  Debian GNU/Linux 12 (bookworm)
- Data protection
  Snapshot schedules
- Networking
  1 network interface
- Observability
  Install Ops Agent
- Security
- Advanced

Operating system and storage

Name	instance-20250403-004151
Type	New balanced persistent disk
Size	10 GB
Snapshot schedule	default-schedule-1
License type	Free
Image	Debian GNU/Linux 12 (bookworm)

The selected image requires you to install an NVIDIA CUDA stack manually. To skip manual setup, click "Switch Image" below to use a GPU-optimized Debian OS image with CUDA support at no additional cost.

Switch image

Change

Additional disks

+ Add new disk | + Attach existing disk | + Add local SSD

Container

Deploy a container image to this VM instance

Deploy container

Monthly estimate

$207.95

That's about $0.28 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
1 vCPU + 3.75 GB memory	$40.15
1 NVIDIA T4	$255.50
10 GB balanced persistent disk	$1.00
Use discount	-$88.69
Logging	Cost varies
Monitoring	Cost varies
Snapshot schedule	Cost varies
Total	$207.95

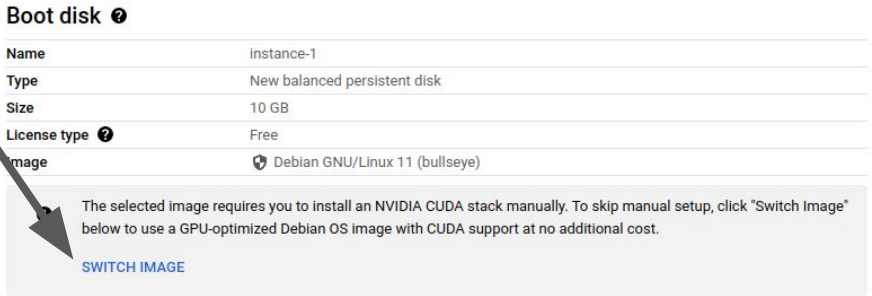
Compute Engine pricing

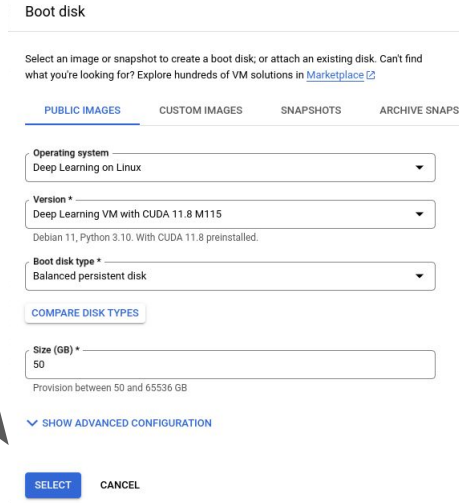
Cloud Operations pricing

Less

4. Create an instance with attached GPUs

- Choose Boot disk and image





- Change firewall setting
  - Select allow HTTP and HTTPS traffic, so you can install packages and connect to GitHub server
- Click ”Create”

Search (/) for resources, docs, products

Confidential Computing is disabled on this VM instance

ENABLE

Container

Deploy a container image to this VM instance

DEPLOY CONTAINER

Boot disk

Name	instance-1
Type	New balanced persistent disk
Size	10 GB
License type	Free
Image	Debian GNU/Linux 11 (bullseye)

CHANGE

Identity and API access

Service accounts

Service account

Boot disk

Select an image or snapshot to create a boot disk; or attach an existing disk. Can't find what you're looking for? Explore hundreds of VM solutions in Marketplace

PUBLIC IMAGES | CUSTOM IMAGES | SNAPSHOTS | ARCHIVE SNAPSHOTS | EXISTING DISKS

Operating system
Deep Learning on Linux

Version *
Deep Learning VM with CUDA 11.8 M115

Debian 11, Python 3.10. With CUDA 11.8 preinstalled.

Boot disk type *
Balanced persistent disk

COMPARE DISK TYPES

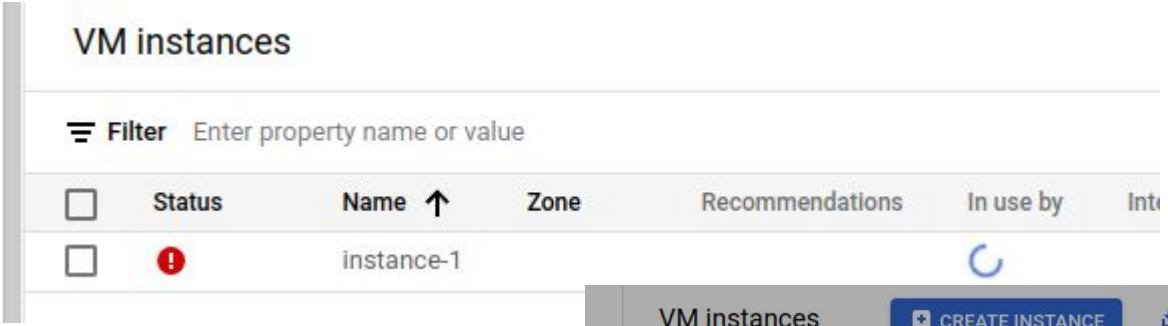
Size (GB) *
50

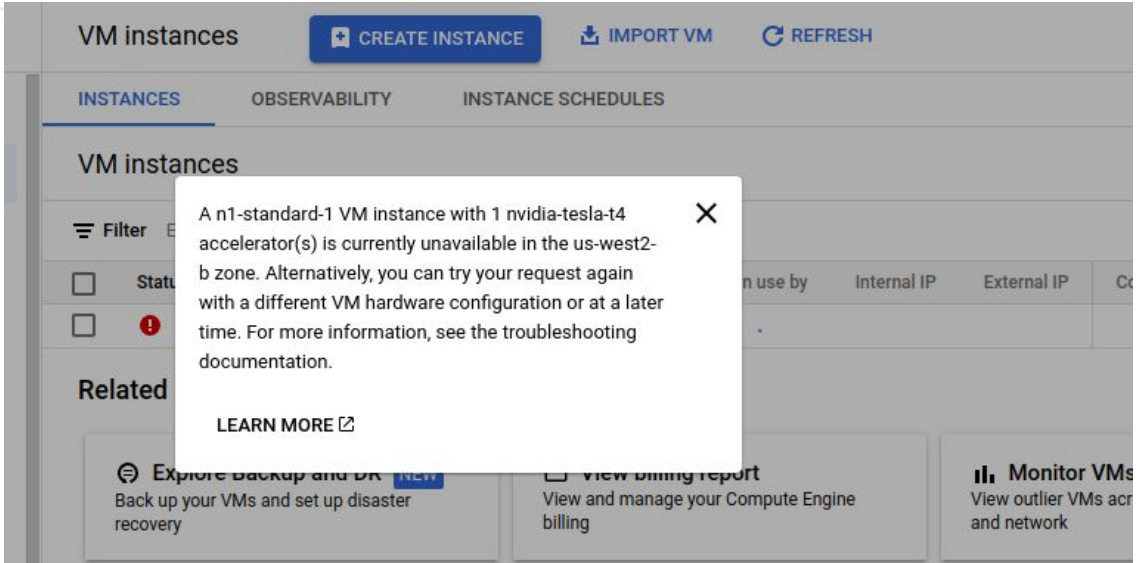
Provision between 50 and 65536 GB

SHOW ADVANCED CONFIGURATION

SELECT CANCEL

Note: If you have difficulty obtaining a machine, try using another region instead

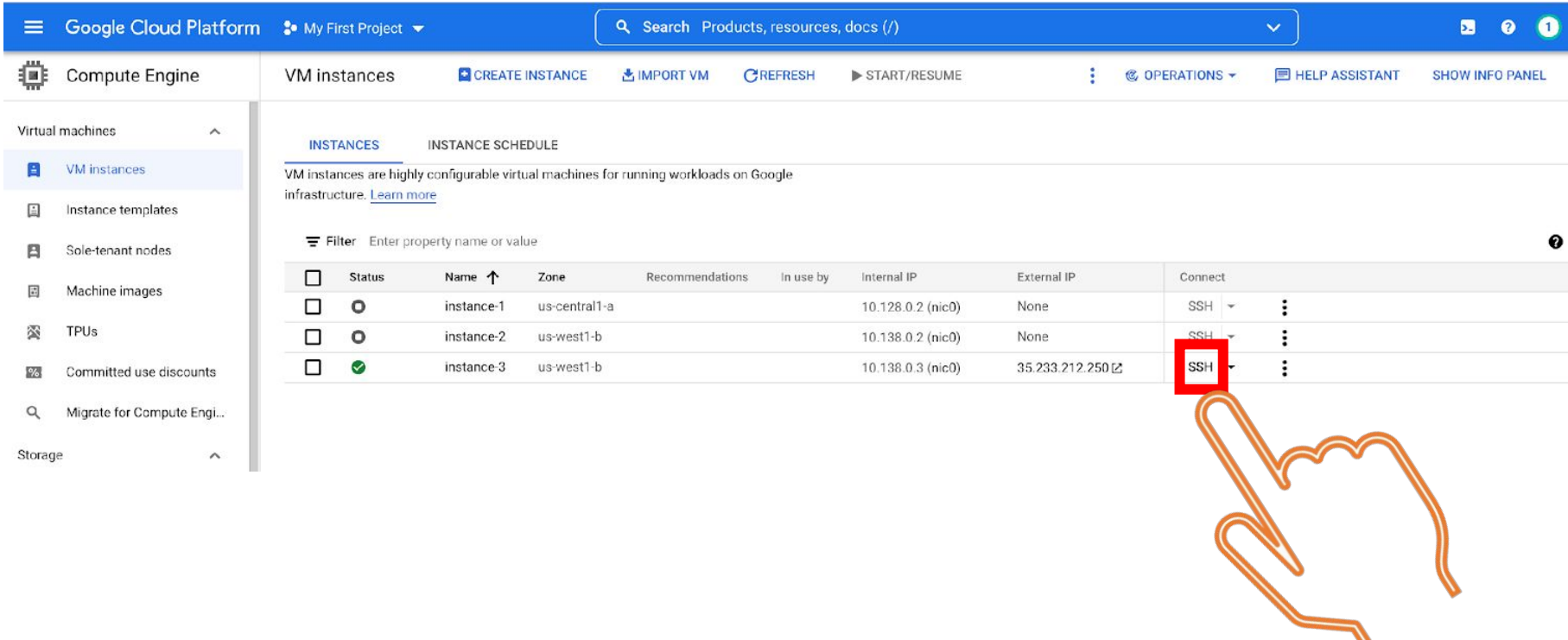




5. Install GPU driver

- If you choose the image with CUDA, your GPU driver will be installed automatically when you first login your machine
  - SSH into your machine in the Google Cloud portal (you have to login using your admin account to install the driver)
  - Input “y” when it prompts “Would you like to install the NVIDIA driver? ”

5. Install GPU driver



5. Install GPU driver

SSH: instance-3 @ commanding-day-337807

https://ssh.cloud.google.com/projects/commanding-day-337807/zones/us-west1-b/instances/instance-3?authuser=

```
==========================================

Version: common-cu113.m87
Based on: Debian GNU/Linux                                  d64 x86_64\n
)

Resources:
 * Google Deep Learning Platform StackOverflow: https://stackoverflow.com/questi
ons/tagged/google-dl-platform
 * Google Cloud Documentation: https://cloud.google.com/deep-learnin
 * Google Group: https://groups.google.com/forum/#!forum/google-dl-p

To reinstall Nvidia driver (if needed) run:
sudo /opt/deeplearning/install-driver.sh
Linux instance-3 4.19.0-18-cloud-amd64 #1 SMP Debian 4.19.208-1 (2021-09-29) x86
_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.

This VM requires Nvidia drivers to function correctly.   Installation takes ~1 minute
Would you like to install the Nvidia driver? [y/n] y
```

Please consider adding the IAP-secured Tunnel User IAM role to start using Cloud IAP for TCP forwarding for better performance. Learn more Dismiss

This is your username

imamingyu@instance-3: ~

https://ssh.cloud.google.com/projects/commanding-day-337807/zones/us-west1-b/instances/instance-3?authuser=0&hl=en...

```
Uncompressing NVIDIA Accelerated Graphics Driver for Linux-x86_64 460.73.01.....
.........................................................................................
.........................................................................................
.........................................................................................
.........................................................................................
.........................................................

WARNING: The nvidia-drm module will not be installed. As a result, DRM-KMS will not function with this
         installation of the NVIDIA driver.


                                   d to guess the X library path '/usr/lib64' and X module path
                                   these paths were not queryable from the system.  If X fails to find the NVIDIA
                                   stall the `pkg-config` utility and the X.Org SDK/development package for your
         distribution and reinstall the driver.

Nvidia driver installed.
imamingyu@instance-3:~$ nvidia-smi
Fri Jan 21 01:20:24 2022
+-----------------------------------------------------------------------------+
| NVIDIA-SMI 460.73.01    Driver Version: 460.73.01    CUDA Version: 11.2     |
|-------------------------------+----------------------+----------------------+
| GPU  Name        Persistence-M| Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp  Perf  Pwr:Usage/Cap|         Memory-Usage | GPU-Util  Compute M. |
|                               |                      |               MIG M. |
|===============================+======================+======================|
|   0  Tesla K80           Off  | 00000000:00:04.0 Off |                    0 |
| N/A   36C    P0    67W / 149W |      0MiB / 11441MiB |      0%      Default |
|                               |                      |                  N/A |
+-------------------------------+----------------------+----------------------+
|   1  Tesla K80           Off  | 00000000:00:05.0 Off |                    0 |
| N/A   52C    P0    74W / 149W |      0MiB / 11441MiB |    100%      Default |
|                               |                      |                  N/A |
+-------------------------------+----------------------+----------------------+

+-----------------------------------------------------------------------------+
| Processes:                                                                  |
|  GPU   GI   CI        PID   Type   Process name                  GPU Memory |
|        ID   ID                                                   Usage      |
|=============================================================================|
|  No running processes found                                                 |
+-----------------------------------------------------------------------------+
imamingyu@instance-3:~$
```

5. Install GPU driver

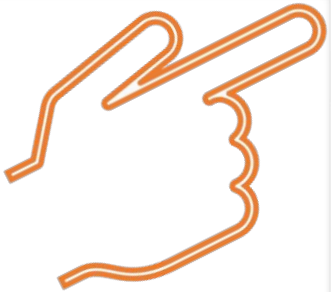
- Verify the GPU driver is installed
  - Type “nvidia-smi” command, you should see this if the driver is installed successfully

5. Install GPU driver

- Otherwise you could following steps in the following link
  - Installing GPU drivers | Compute Engine Documentation | Google Cloud
- We need to install
  - NVIDIA driver
  - CUDA toolkit
  - CUDA runtime

Turn off your machine when it’s not using

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
	instance-1	us-central1-a			10.128.0.2 (nic0)	None	SSH
	instance-2	us-west1-b			10.138.0.2 (nic0)	None	SSH
	instance-3	us-west1-b			10.138.0.3 (nic0)	35.233.212.250	SSH



- Start/Resume
- Stop
- Suspend
- Reset
- Delete
- View network details
- Create new machine image
- View logs
- View monitoring

- So you can save some credit

~~Setup Google Colab using VM as Backend~~

Google Colab is now deprecated from VMs.
More Info

Alternatives:

- Connect to VM with SSH and VSCode (recommended)
  - Develop code using a local VS Code editor | Cloud Workstations | Google Cloud
  - Remote Development using SSH
- gcloud CLI and JupyterLab
  - Quickstart: Create a Deep Learning VM instance by using the gcloud CLI
  - Connect to JupyterLab | Deep Learning VM Images | Google Cloud

Tips for Experiments on Remote Machine

Connect to your instance

- If you would like to connect to your machine using terminal directly, instead of using the browser-based ssh window
- Create key (Detailed tutorial: How to Use SSH Public Key Authentication – ServerPilot)
  - Using command ssh-keygen
  - You will keep the private key (for example id_rsa) in your local computer
- Add key
  - Add public key (like id_rsa.pub) to your Google Cloud instance setting
  - Click into your instance, click “Edit” in the top navigation bar, find “SSH key”, click “Add Item”, enter your public SSH key content there
- Connect your remote instance from your local terminal

  ssh –i key_path username@external_ip_address
- Connecting to Linux VMs using advanced methods | Compute Engine Documentation | Google Cloud

Access file and coding remotely

- You will need to edit code and run the updated codebase with new implementation
- Choice 1: VS Code
  - Developing on Remote Machines using SSH and Visual Studio Code
- Choice 2: PyCharm
  - Getting started with remote development | PyCharm (jetbrains.com)
- Choice 3: transfer files by scp/sftp
  - Using scp/sftp to transfer file/code from your local machine to the remote machine

Monitor and specify GPU usage

- Check whether your job is running on GPU, memory usage, job ID etc
  - nvidia-smi
- Specify which GPU(s) to use
  - export CUDA_VISIBLE_DEVICES="0”
  - export CUDA_VISIBLE_DEVICES="0,1,2”
  - export CUDA_VISIBLE_DEVICES=""

Run experiments in background

- Use tmux to run your job in background, so your job can continue running if your ssh session broke
- tmux new -s exp1
  - Create a new tmux session
- control + b, then press d
  - Exit the session
- tmux a –t exp1
  - Enter the session exp1 again
- tmux ls
  - See all active sessions

Install environment and run code

Install miniconda

>>> wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh

>>> sh Miniconda3-latest-Linux-x86_64.sh

Create conda environment

>>> conda create -n ece239as python==3.xx

>>> conda activate ece239as

Install dependencies needed

>>> conda install pip

>>> pip3 --no-cache-dir install torch==1.10.1+cu113 torchvision==0.11.2+cu113 torchaudio==0.10.1+cu113 -f https://download.pytorch.org/whl/cu113/torch_stable.html